Alexandria Engineering Journal 128 (2025) 223–230

Contents lists available at ScienceDirect

Alexandria Engineering Journal

journal homepage: www.elsevier.com/locate/aej

Original Article

A hybrid blockchain and AI-based approach for attack protection to secure internet of medical things networks

Hadeel Alsolai [a], Fahd N. Al-Wesabi [b], Ali M. Al-Sharafi [c], Asma Alshuhail [d], Abdulbasit A. Darem [e,*], Bandar M. Alghamdi [f], Fouad Shoie Alallah [g], Abdulrahman Alzahrani [g]

[a] Department of Information Systems, College of Computer and Information Sciences, Princess Nourah Bint Abdulrahman University, Saudi Arabia
[b] Department of Computer Science, Applied College at Mahayil, King Khalid University, Saudi Arabia
[c] Department of Computer Science and Artificial Intelligence, College of Computing and Information Technology, University of Bisha, Bisha 67714, Saudi Arabia
[d] Department of Information Systems, College of Computer Sciences & Information Technology, King Faisal University, Saudi Arabia
[e] Center for Scientific Research and Entrepreneurship, Northern Border University, Arar 73213, Saudi Arabia
[f] Department of Information Technology, Faculty of Computing and Information Technology, King Abdulaziz University, Jeddah 21589, Saudi Arabia
[g] Department of Computer Science and Engineering, College of Computer Science and Engineering, University of Hafr Al Batin, Saudi Arabia

ARTICLE INFO

Keywords:
Internet of medical things (IoMT)
Blockchain security
Federated learning
AI-driven anomaly detection
Cyber threat mitigation

ABSTRACT

The Internet of Medical Things (IoMT) has revolutionized healthcare by enabling real-time monitoring, remote diagnosis, and seamless data exchange among medical devices and providers. However, the interconnected and distributed nature of IoMT systems exposes them to significant cybersecurity threats, including data breaches and unauthorized access. This study aims to develop a secure and scalable framework to safeguard IoMT environments using blockchain technology, federated learning, and dynamic consensus algorithms. The proposed approach integrates artificial intelligence to detect threats and employs federated learning to maintain patient privacy while training models collaboratively. Experimental evaluation of the framework shows improved data security, reduced response times, and enhanced system trustworthiness. The results suggest that the integration of blockchain and AI-driven mechanisms offers a robust solution to IoMT cybersecurity challenges, paving the way for more reliable and efficient healthcare systems.

1. Introduction

Through Internet of Things technology, healthcare providers and medical devices can monitor and diagnose health conditions in real time, as well as seamlessly exchange data back and forth. A growing number of Internet of Medical Things (IoMT) devices are revolutionizing healthcare with real-time monitoring, remote diagnosis, and seamless data exchange[1]. It is still highly vulnerable to cyber-attacks, such as breaches and unauthorized access, despite its interconnected nature. Due to the dynamic and distributed nature of IoMT networks, traditional security mechanisms may not be able to provide comprehensive protection. Adaptable Blockchain Frameworks are proposed in this article to enhance IoMT's security and resilience in the face of these challenges. Cyber threats are evolving on a daily basis, which is why adaptive mechanisms such as AI-driven anomaly detection and dynamic consensus algorithms are necessary. The healthcare industry is experiencing rapid growth in the IoMT sector [2]. Using them, you can capture, create, analyze, and transfer information. The IoMT infrastructure relies on connected devices. Several critical diseases can be monitored in real-time using linked IoMT devices, saving the lives of many patients. As a result of the Internet of Things, doctors are in short supply in the healthcare sector due to the automation of all health-related functions, including data collection, predictive analysis, and report generation[3]. Through the integration of digital and physical worlds, the IoMT improves patient safety and precision in diagnosis and care [4]. Patients and clinicians will benefit greatly from the integration of medically related devices. In real-time, the IoMT can help doctors and nurses diagnose more quickly and treat more accurately, enhancing patient safety and enhancing patient behaviour[5]. Incorporating medically related devices will profoundly affect the outcomes of patients and clinicians [6]. These volumes of data are difficult to analyze, store, and manage with traditional methods. The success of

* Corresponding author.
E-mail address: basitdarem@nbu.edu.sa (A.A. Darem).

https://doi.org/10.1016/j.aej.2025.05.054
Received 9 March 2025; Received in revised form 5 May 2025; Accepted 14 May 2025
Available online 28 May 2025
1110-0168/© 2025 The Authors. Published by Elsevier B.V. on behalf of Faculty of Engineering, Alexandria University. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

IoMT-enabled healthcare systems relies on distributed data storage. (Fig. 1)

Embracing the world at a rapid rate is the IoT, a revolutionary technology. Combined with AI, blockchain, and 5 G, IoT enhances personal experiences by providing contextual connectivity [7], [8], [9]. As a result of resource constraints, IoT devices rely on traditional security mechanisms like passwords, which can be easily hacked. Since remote access is insecure, IoTs are susceptible to hacks [10]. E-health allows electronic monitoring, maintenance, and improvement of patient health. As IoT technology advances, the healthcare sector is rapidly transforming its hospital-centric services into home-centric ones. Security and privacy in healthcare are founded on trust, which is the cornerstone of IoTs in healthcare. Although IoT devices are interconnected and rely on traditional security mechanisms, they rely on centralized architectures that are incompatible with IoT devices when it comes to trust.

Apart from being a new technology, blockchain also has many inherent characteristics, including decentralized applications, decentralized trust, transparency, and immutability [11]. Blockchain's centralized trust and immutability make it ideal for IoT applications, especially in healthcare [12], [13]. As part of smart diagnosis, disease profiling, vitals management, and AI-driven disease detection, healthcare devices record sensitive patient data that can be analyzed to provide smart solutions. A smart ventilator can fail in an intensive care unit, for example, causing a patient's death. Similarly, insiders selling sensitive healthcare data on the black market for personal gain constitute a breach of trust. Data leaks or mishandling are punishable by severe fines under HIPAA and GDPR to protect patient privacy. IoT healthcare devices are usually designed without security considerations since manufacturers don't comply with standards. Therefore, it is imperative to have an adequate security system that is both varied and compliant with modern health requirements.

IoMT-enabled healthcare networks can benefit from blockchain's cryptographic properties and decentralization, making it a suitable technology for authentication and access services. Among the peers of a blockchain, cryptographic hashes are distributed, which link each block time stamped and linked by cryptographic hashes [14], [15]. Blockchain technology was designed primarily to facilitate money exchanges between two parties in the beginning. Security professionals around the world are using the blockchain in IoMT to protect security and privacy better. Authentication, authorization, and storage are some of the many fields where blockchain is increasingly used due to its reliability, tamperproof nature, and fault tolerance [16]. A blockchain-based smart contract can also be used to control IoMT devices in the healthcare sector. Blockchain technology and IPFS clusters enable IoMT systems to manage decentralized infrastructures, trust, integrity, authenticity, privacy, and security [17]. The Internet of Medical Things enables the sharing of healthcare data between medical devices and people through wireless connectivity. Healthcare and utility costs are increasing due to population growth and new technology. It can improve diagnosis accuracy, reduce treatment costs, and reduce complications, among other benefits [18]. Cloud servers are used to track the success of patient data collection in healthcare. Cloud servers provide measurement and storage resources to measure the intensity of a quantitative evaluation. The integration, exchange, and transfer of data to different devices for processing create medical security challenges, such as data protection. Further, the openness and protection of data are inadequate, which creates legal and compliance issues [19]. Blockchain technology is used to manage and acquire data securely. Traditional security mechanisms such as centralized access control and encryption are often inadequate in addressing the complex, decentralized infrastructure of IoMT. Consequently, there is a critical need for robust, adaptive, and scalable cybersecurity solutions specifically tailored for the IoMT ecosystem. Existing centralized security mechanisms are not scalable or resilient enough for distributed IoMT environments. Ensuring data privacy while enabling efficient machine learning remains a significant challenge. This research proposes a comprehensive security framework that leverages blockchain technology for decentralized trust, federated learning for privacy-preserving data analysis, and dynamic consensus algorithms to improve performance and reliability. The fusion of these technologies addresses key vulnerabilities in IoMT systems while maintaining high efficiency and compliance with privacy standards.

Main Contributions:

- A novel integration of blockchain, federated learning, and AI-based threat detection tailored for IoMT.

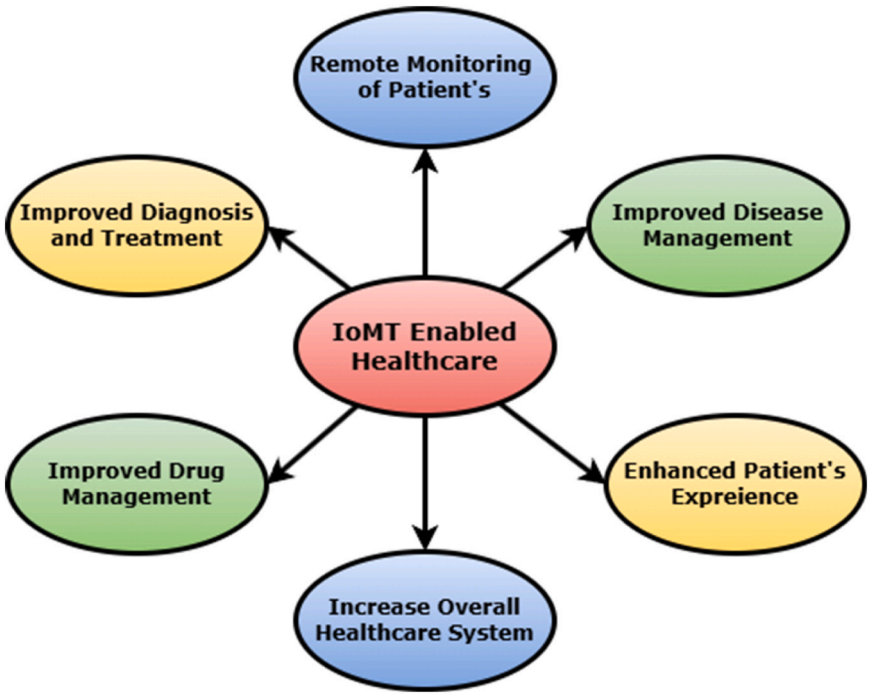

Fig. 1. Benefits of IoMT in healthcare system.

- A privacy-preserving federated learning approach that ensures patient data confidentiality.
- Experimental validation demonstrating improved security, reduced latency, and scalability of the proposed framework.

Paper Organization: The rest of the paper is organized as follows: 2 reviews related work and highlights the research gaps. 3 outlines the proposed system architecture and methodology. 4 presents the experimental setup and results. 5 discusses the implications and limitations of the work. Finally, Section 6 concludes the paper and suggests future research directions.

2. Related Work

Studies measuring and comparing health outcomes have been conducted for many years. Despite this, since IoMT is relatively new, there is very little information about how to evaluate and recommend its security. There has been a long history of measuring and comparing health. The newness of IoMT makes it difficult to recommend and evaluate security in IoMT applications. A review of previous work in this area is presented in the following paragraph, as well as their challenges and shortcomings. In some cases, these efforts are aimed solely at patient monitoring systems and do not take into account other aspects of healthcare [20], [21]. In addition to research on IoMT security features, there is no public availability of these tools, which might be difficult for novice users to use. Furthermore, some IoMT evaluation tools draw on hypothetical and generalized security recommendations. Data privacy is not mentioned, but they focus mostly on the physical security of IoMT [22], [23]. Several other techniques are proposed that do not require users to provide security information [24]. Additionally, some researchers have developed security evaluation criteria using IoT architectures. As a result of its interconnected architecture, there is no clear boundary ensuring the security of IoMT [25].

Technology advances are leading to massive amounts of data being generated that can't be stored and processed by traditional systems today. Due to these challenges, a cloud-based centralized system is used to manage and process data. Using cloud-based healthcare systems, the Author [26] proposed privacy-preserving methods for the healthcare system. The data inputs and outputs using this approach are secure, but the system cannot be trusted. Moreover, IoMT-enabled healthcare systems face the challenge of storing data generated by IoMT. At present, all data storage is centralized, such as on cloud infrastructures, and third parties authenticate the data [27], [28]. This paper presents some work on storing and authenticating data. A model for the deployment of healthcare systems was proposed by the Author [29]. The authors of this paper analyze security risks associated with service providers and clouds. Moreover, they stated that information management systems could be implemented on a wide range of clouds. Their report, however, did not explain how medical records could be protected against unauthorized access using cryptographic algorithms and access control techniques. A cloud-assisted gateway architecture was also presented by the Author [30] to facilitate patient access to cloud health resources. It's presented a blockchain-driven approach for secure data transmission. While the study effectively demonstrated enhanced data integrity and tamper resistance, it did not incorporate privacy-preserving machine learning techniques or dynamic consensus mechanisms that adapt to system changes [1], [31].

An Author [32] designed a system for managing health records that enables patients and doctors to access records with key control methods. There is no discussion in this paper on how to keep data private and secure. An authentication and authorization approach based on proof-based authentication was described in [33]. The focus was primarily on data integrity and authentication, with little attention being paid to other security measures, such as access and audit trails. Several technologies and methodologies are incorporated into the IoMT to enhance healthcare efficiency, accessibility, and quality. In the IoMT architecture, interconnected devices and communication protocols facilitate seamless monitoring, diagnosis, and treatment. In Table 1, it is demonstrated how this architecture has impacted modern healthcare.

3. Proposed Methodology

The proposed framework aims to secure real-time data transfer in IoMT environments by integrating artificial intelligence, federated learning, and blockchain technology. This hybrid approach addresses multiple layers of cybersecurity while preserving data privacy and ensuring system scalability. The framework consists of four main components: IoMT devices, edge nodes, a federated learning engine, and a blockchain network. IoMT devices collect patient data and transmit it to edge nodes for local processing. The federated learning engine allows decentralized training of machine learning models without sharing raw data, ensuring patient privacy. The blockchain network provides a tamper-resistant ledger for storing model updates, access logs, and consensus decisions.

Federated learning enhances privacy by keeping data localized. However, it is vulnerable to poisoning attacks, where compromised nodes submit malicious updates to corrupt the global model. To counter this, the framework implements robust aggregation techniques and anomaly scoring to detect and filter suspicious updates. Additional defense include differential privacy techniques and adversarial training to strengthen model resilience against manipulation. This comprehensive methodology balances real-time data security with system efficiency, making it well-suited for the complex and sensitive environment of IoMT systems.

A weights w_G is initially assigned to each IoMT device, ensuring that each device receives the parameters associated with the global model. In this way, each device will have a consistent model at the start of its localized training, thereby maintaining model coherence between them. In the following step, every IoMT device starts utilizing its specific patient dataset to start the localized training process. As shown, the entire training process occurs on the device, ensuring that raw patient data remains on the device. A weight update, or Δw_{L_i}, is calculated after the training to reflect the difference between the local and global models, according to each device. Keeping patient data private is our priority, so only the patient's weight update is sent to our central server. Δ_{w_G} is a weight update composed of the average weights of all participating devices. When all participating devices submit weight updates, the central server computes this update. When all IoMT devices are included in the global model, accuracy and generalizability increase. Hosting the global model is the primary function of the central server. w_G denotes the random weights used to initialize G, the global model

$$G : X \rightarrow Y, wirh\ initial\ weights W_g \tag{1}$$

Using this global model, IoMT devices may be adapted for localized training. D_i represents the subset of data stored in each IoMT device. The device's localized models L_i are trained with G. Mathematically; training can be described as follows:

$$L_i : X_i \rightarrow Y_i, where D_i \cap X_i \times Y_i \tag{2}$$

Every L_i optimizes its weights w_{L_i} according to D_i Using several iterations. Δw_{L_i} (representing the changes from w_G) is the only information received by the central server.

$$\Delta w_{L_i} = w_{L_i} - w_G \tag{3}$$

While ensuring minimal data transfer, raw patient information is protected. Using the aggregated data from all participating devices, the central server refines. W_G. In a simple aggregation, all updates are averaged:

$$\Delta w_G = \frac{1}{N} \sum_{i=1}^{N} \Delta w_{L_i} \tag{4}$$

Table 1
IOMT architecture in practice.

IoMT Model	Description	Ref.
Remote patient monitoring	Patients with chronic diseases often wear wearable devices to monitor vital signs like heart rate, blood pressure, glucose level, and exercise level.	[34], [35]
Monitoring pregnancy	As part of prenatal healthcare, self-monitoring in the home is an essential component. As part of pregnancy, pregnant women can monitor and manage maternal health indicators using wearable technologies, including fetal monitors and multifunctional health screening tools. These include blood pressure, fetal blood sugar, fetal heart rate, oxygenation, pulse, lipids, and electrocardiograms.	[36], [37]
Personalized healthcare	Using data-driven insights, the IoMT provides patients and caregivers with customized treatment plans, and smart pillboxes alert them when they need to take prescribed medications. With the help of wearable technology, you can track lifestyle behaviours (such as sleep and activity) and make changes according to the results.	[38]
Optimization of hospital workflow	The IoMT automates and streamlines clinical processes in healthcare facilities. Equipment such as wheelchairs and ventilators can be tracked using RFID systems.	[39]
Early warning systems and real-time diagnostics	AI algorithms in IoMT detect irregularities and alert people so they can take action immediately. As soon as an arrhythmia or heart attack occurs, emergency services are automatically alerted. People wear Electroencephalography (EEG) and ECG scanners to detect abnormalities in their hearts or brains.	[40]
Therapy and rehabilitation	The IoMT helps rehabilitation programs by utilizing interactive and adaptive technologies. It is possible to track progress and adjust regimens in real time using wearable technology. After a stroke or brain injury, people are able to recover using brain-machine interfaces. It is possible to deliver targeted therapy using transcutaneous electrical nerve stimulation units.	[41]
Drug delivery and medication systems	A sophisticated medication delivery system is integrated with the IoMT to ensure precise dosages and times. Based on the results of ongoing blood glucose tests, insulin pumps adjust the insulin dosage automatically. The use of inhaler sensors tracks and reports medication adherence for asthmatics and people with chronic obstructive pulmonary disease. Cancer medications and chronic pain medications can be released through implantable drug systems in a controlled and programmable manner.	[40]
Systems for responding to emergencies	The IoMT enhances the efficiency of emergency response in emergency scenarios. Seniors and those in danger can wear wearable panic buttons to receive instant alerts. A hospital receives real-time patient data from ambulances equipped with IoMT. Sensors monitor risks and casualties associated with the IoMT.	[40]
Clinical trials and research	Data collection in various real-world contexts is made easier with IoMT devices. As a result, self-reports do not need to be as accurate as real-time data. The Internet of Things facilitates remote participation, enhancing candidates' chances of being selected. Long-term, comprehensive datasets result from ongoing monitoring.	[42]
Epidemic management and public health	In addition to providing extensive health monitoring, the IoMT helps with reaction planning. Infectious disease exposure can be monitored with wearable technology, and health data can be collected using sensors. Follow-ups are planned, and digital platforms guarantee compliance.	[43], [39]
Home medical care	Remote monitoring is also possible through the IoMT, in addition to receiving medical care from a home-based clinic. Doctors use blood pressure monitors, thermometers, and glucose meters to measure blood pressure.	[43], [39]

A total of N devices will be included in this calculation. A new set of weights is then applied to the model:

$$W_{G_{new}} = W_G + \Delta W_G \tag{5}$$

The global model G is continuously refined so that insight from all devices can be incorporated without exposing individual information.

3.1. Blockchain Authentication and Verification Model

Digital signatures using this method (ECDSA) are encrypted using elliptic curves [44]. Sending a message with an elliptic curve signature to a receiver is an example of this. To achieve an agreement, both sides must define a set of parameters and express them as a $(CURVE, G, n)$. When two elliptic *curves* are joined together, the dot product is calculated using the geometric equation of the elliptic curve G, N as its multiplicative order, and $N_G = 0$ as its zero point. The integration of cryptographic techniques enhances data security, but the framework must also address the complexities of key management, particularly in distributed IoMT environments. Challenges such as secure key generation, distribution, rotation, and revocation require robust solutions to prevent unauthorized access and ensure scalability. Exploring lightweight key management protocols or decentralized approaches like blockchain-based key distribution can help mitigate these issues effectively.

A message is sent once a private key and a public key have been created. $[1, n-1]$ is a range of random numbers for the private key.

$$d_A = rand(1, n-1) \tag{6}$$

A public key is derived from a private key and a base point by doing the elliptic curve dot product:

$$Q_A = d_A \times G \tag{7}$$

3.1.1. Signature algorithm

m is the signer's signature. Users' names and attributes are included in this scheme. Follow these steps to complete the process:

Step 1: Compute $e = h(m)$, anywhere $m = (UID|h(pswd))$;

Step 2: An integer z is calculated by dividing binary e by its leftmost bit (the highest L bit), where L is n's binary length.

Step 3: Randomly select a number k from the range $[1, n-1]$;

Step 4: Using the elliptic curve, calculate a point:

$$(x_1, y_1) = k \times G \tag{8}$$

Step 5: Using step2, calculate r and recalculate in step3 if $r == 0$

$$r = x_1 \mathrm{mod} n \tag{9}$$

Step 6: When $s == 0$, recalculate step 3 if $s == 0$

$$s = k^{-1}(z + rd_A)\mathrm{mod} n \tag{10}$$

Step 7: A digital signature consists of a produced (r,s) code.

3.1.2. Authentication algorithm

Receivers will also receive public keys along with signatures. A public key needs to be authenticated first, then the signature (r, s) must

be authenticated

Authentication of public keys

Step 1: To prevent null points O from exceeding the limit value, the coordinates of public key Q_A need to be valid

Step 2: It is verified that points on an elliptical curve are associated with Q_A coordinates using a public key;

Step 3: Formulation (11) must be able to be worked out, i.e. the multiplicative order n must not be a point product of the public key.;

$$n \times Q_A = O \tag{11}$$

Authentication of signature files

Step 1: To authenticate, r and s must be integers between $[1, n-1]$; otherwise, the authentication will fail.

Step 2: Compute $e = h(m)$;

Step 3: Based on the highest L bit of e, calculate z

Step 4: The next step is to calculate w by using

$$w = s^{-1} mod n \tag{12}$$

Step 5: Then u_1 then u_2

$$u_1 = zw\ mod n, \quad u_2 = rw\ mod n \tag{13}$$

Step 6: To authenticate, you must calculate (x_1, y_1) as an elliptical point

$$(x_1, \quad y_1) = u_1 \times G + u_2 \times Q_4 \tag{14}$$

Step 7: A valid authentication requires the following formula; otherwise, a failure occurs.

$$r \equiv (x_1 \mathrm{mod} n) \tag{15}$$

There are two conditions for a successful verification: first, the verification information must be accurate, and second, the consensus node must reach the minimum number of verifications,

$$t \geq t_0, \quad t_0 = 2f + 1 \leq n \tag{16}$$

A minimum of three correct nodes (t_0) is required in PBFT $t_0 = n - f \geq 2f + 1$, for example, if $f = 1$. A successful authentication results in the endorsement node returning success information to the client, while a failed authentication results in the return of failure information.

3.2. Blockchain Authentication and Network Topology

The blockchain technology enhances the security and transparency of IoMT systems, its inherent transaction validation latency poses a challenge for real-time healthcare applications where immediate data processing is critical. The proposed framework addresses this by evaluating the impact of various consensus mechanisms on latency and introducing optimization strategies such as asynchronous validation, batching of non-critical transactions, and the use of pre-approved smart contract triggers for urgent operations. These measures aim to maintain the required responsiveness for time-sensitive clinical tasks without compromising data integrity or system trustworthiness. As part of the Blockchain Authentication Scheme, each role server must be independently deployed, as well as the blockchain network and identity authentication services. There are two types of nodes in a blockchain network: endorsers and peers. The authentication service deploys a server to register and issue certificates. A diagram showing the entire topology of the Blockchain Authentication Scheme is shown in Fig. 2. Among the blockchains established in the Blockchain Authentication Scheme are one ordering service node (OSN) and two organizations (orgs). Certificate Authorities (CAs) and peer nodes are included in each organization. In the blockchain, every node is a committer. Whenever a registration request is submitted, the system enrols an administrator, who is already registered with other tools for each CA of the organization, for the purpose of registering, enrolling, and deregistering users.

Blockchain Authentication Scheme consists of a client and a server. User interfaces and data collection are the client's responsibility. Servers deploy independently, and Solo is used for consensus. As a result, the HyperLedgerFactory1.4 is deployed on the server to manage requests, data, and transactions. The effectiveness of the cryptographic techniques used in the proposed model should be benchmarked against current IoMT encryption standards, such as AES, ECC, and TLS protocols. This comparison ensures that the framework meets industry-accepted levels of confidentiality and integrity while remaining computationally efficient for resource-constrained devices. Benchmarking also helps identify potential trade-offs between security strength and system performance.

This approach enhances fault tolerance by ensuring that the failure of

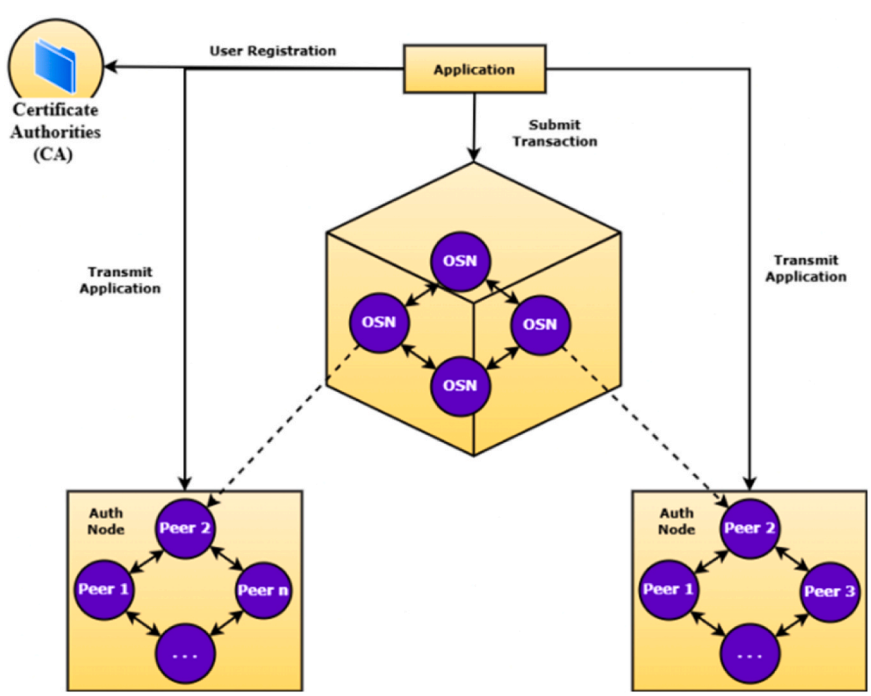

Fig. 2. The topology of the Blockchain Authentication Scheme.

any single node does not halt the training process. Additionally, backup nodes are designated to take over responsibilities in case of unexpected dropouts, and a lightweight consensus algorithm ensures agreement on node selection and task assignments. Together, these strategies enhance system robustness and ensure continuity in model training even in the presence of node failures. The proposed authentication mechanism must also account for the risks posed by stolen or compromised IoMT device identities. Since IoMT devices often operate in unsecured or minimally supervised environments, they are particularly vulnerable to identity spoofing or credential theft. To mitigate this risk, the framework should incorporate dynamic identity verification techniques such as behavioral biometrics, device fingerprinting, or challenge-response protocols that go beyond static credentials. Furthermore, integrating anomaly detection systems can help identify suspicious behavior indicative of compromised devices. These enhancements are essential to ensure that only legitimate and uncompromised devices participate in the federated learning process, thereby preserving data integrity and trust within the system.

4. Result and Discussion

The framework should account for resilience against distributed denial-of-service (DDoS) attacks, which can severely disrupt IoMT networks by overwhelming devices or communication channels. Mitigation strategies may include traffic filtering, rate limiting, and decentralized network architectures that reduce reliance on single nodes. Additionally, integrating anomaly detection to identify and respond to unusual traffic patterns in real time can help contain potential DDoS threats and maintain service availability. In the healthcare sector, blockchain-based solutions improve the quality of data, improve service delivery, and ensure that data remains continuous. Fig. 3 illustrates how internet-based technology can improve patient outcomes through more accurate treatment, faster medical treatments, and more practical treatment methods.

As part of the blockchain framework, cloud servers and personal servers could exchange data securely. As part of the blockchain framework, cloud servers and personal servers could exchange data securely. According to Fig. 4, the response latency ratio is more beneficial than cloud storage and network communication, as well as real-time data interpretation.

Using Fig. 5, we can see that the response time ratio is as follows. Efficacy evaluation is essential to maximizing the value of resources. Optimal health outcomes can be achieved by analyzing how resources are made available within and across healthcare systems. To improve

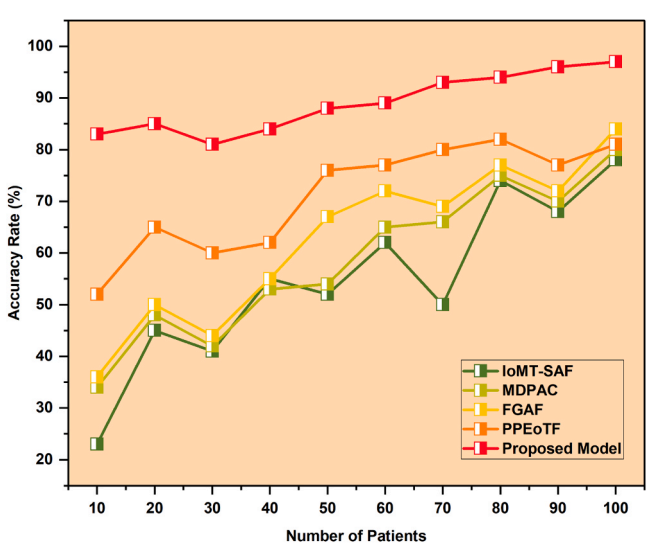

Fig. 3. Overall accuracy ratio.

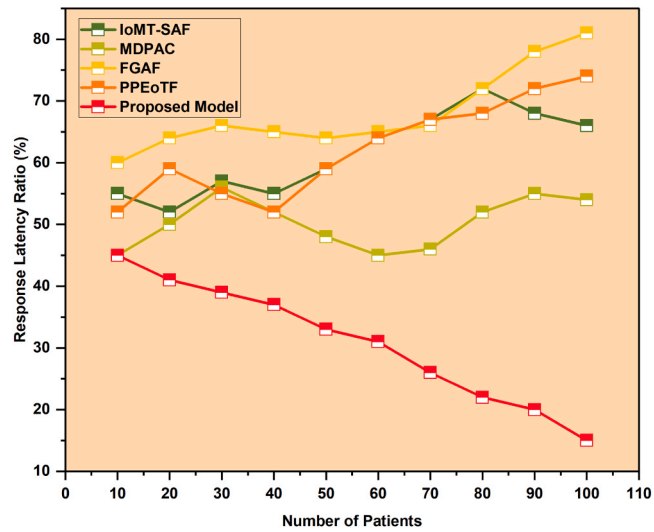

Fig. 4. Relative latency of response.

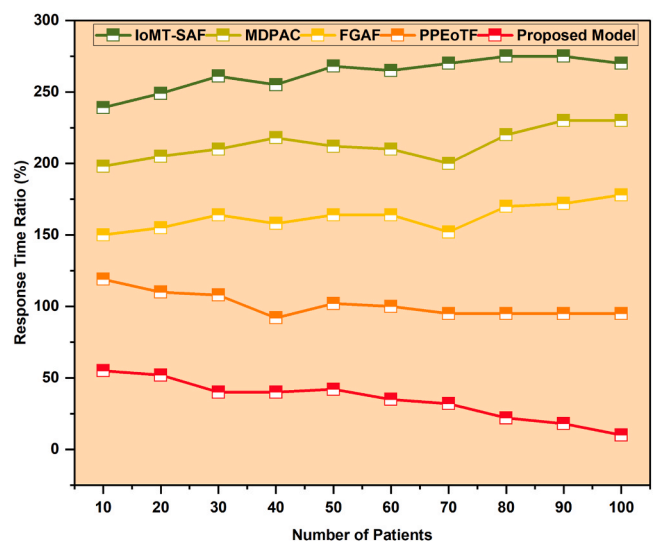

Fig. 5. Response time.

well-being, healthcare services can be considered intermediate commodities when measuring their efficiency.

By evaluating and recording the results, best practices are encouraged and strengthened. A physician's treatment must be respected and redefined through the interpretation of results; Fig. 6 illustrates this concept.

Organizations can use these logs for auditing, identifying areas of concern and improving security measures, where possible. According to Fig. 7, businesses can determine accurate entry points, determine the cause, and assess the harm of an incident by using audit trails.

5. Conclusion

With IoMT's rapid expansion, traditional security mechanisms are having difficulty addressing security and privacy issues. For IoMT networks to be protected, this paper introduces a federated learning and AI-driven adaptive blockchain framework. A decentralized authentication solution, a secure data exchange method, enhanced privacy, and reduced cyberattack risk are provided by the proposed solution. There is an improvement in security, response time, and trustworthiness of the system based on experimental evaluations. This framework uses

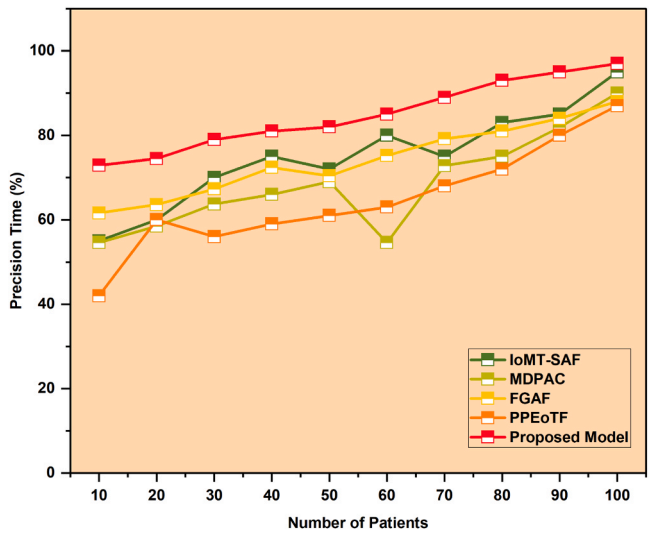

Fig. 6. A ratio of precision.

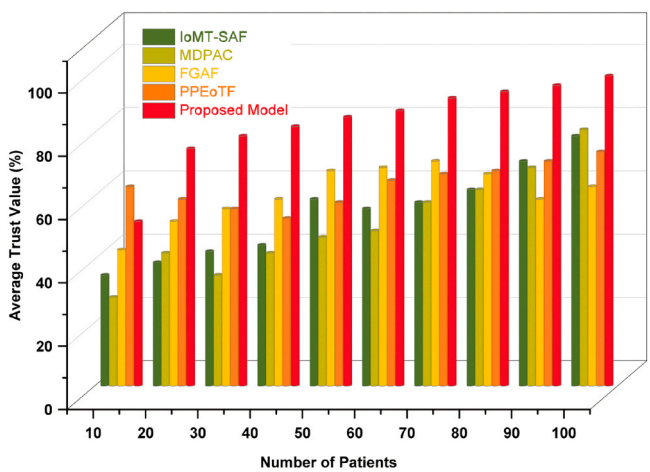

Fig. 7. Average trust value.

blockchain technology, cryptographic techniques, and machine learning to provide scalable and resilient security for IoMT. In the future, research may focus on optimizing computational efficiency and incorporating quantum-resistant cryptographic mechanisms to enhance IoMT security further. A recent study has shown that blockchain has the potential to transform healthcare systems in the future.

CRediT authorship contribution statement

Ali M. Al-Sharafi: Writing – original draft, Validation, Methodology, Conceptualization. **Asma Alshuhail:** Writing – review & editing, Visualization, Validation, Software, Resources, Data curation. **Darem Abdulbasit:** Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology. **Bandar M. Alghamdi:** Writing – review & editing, Writing – original draft, Validation, Conceptualization. **Hadeel Alsolai:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Fahd N. Al-Wesabi:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis. **Fouad Shoie Alallah:** Writing – review & editing, Writing – original draft, Validation. **Alshareef Abdulrhman:** Writing – review & editing, Writing – original draft, Methodology.

Declaration of interest

The authors certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

Acknowledgements

The authors extend their appreciation to the Deanship of Research and Graduate Studies at King Khalid University for funding this work through Large Research Project under grant number RGP2/17/45. Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R303), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia. The authors extend their appreciation to Northern Border University, Saudi Arabia, for supporting this work through project number (NBU-CRP-2025-2903). The authors are thankful to the Deanship of Graduate Studies and Scientific Research at University of Bisha for supporting this work through the Fast-Track Research Support Program. This work was supported by the Deanship of Scientific Research, Vice Presidency for Graduate Studies and Scientific Research, King Faisal University, Saudi Arabia, under project Grant KFU252077.

References

[1] P. Rani, S. Verma, S.P. Yadav, B.K. Rai, M.S. Naruka, D. Kumar, Simulation of the lightweight blockchain technique based on privacy and security for healthcare data for the cloud system, Int. J. E-Health Med. Commun. (IJEHMC 13 (4) (2022) 1–15.

[2] R. Kumar, R. Tripathi, Towards design and implementation of security and privacy framework for Internet of Medical Things (IoMT) by leveraging blockchain and IPFS technology, J. Supercomput 77 (8) (Aug. 2021) 7916–7955, https://doi.org/10.1007/s11227-020-03570-x.

[3] P. Rani, et al., Federated learning-based misbehaviour detection for the 5G-enabled internet of vehicles, IEEE Trans. Consum. Electron. (2023).

[4] D.F.B. Cardoso, Medical Devices Information Systems in Primary Care, Master's Thesis, Universidade NOVA de Lisboa (Portugal), 2022. Accessed: Mar. 03, 2025. [Online]. Available: ⟨https://search.proquest.com/openview/4c4fc23a4d9b19d0fbbe2dc472f79f59/1?pq-origsite=gscholar&cbl= 2026366&diss=y⟩.

[5] P. Rani, U.C. Garjola, H. Abbas, A predictive IoT and cloud framework for smart healthcare monitoring using integrated deep learning model, NJF Intell. Eng. J. 1 (1) (2024) 53–65.

[6] J. Hofdijk, B. Séroussi, C. Lovis, F. Sieverink, F. Ehrler, A. Ugon, Transforming healthcare with the internet of things, in Proceedings of the EFMI special topic conference, 2016. Accessed: Mar. 03, 2025. [Online]. Available: ⟨https://wp-prd.let.ethz.ch/wp-content/uploads/migration/122/2016/12/9781614996330.pdf⟩.

[7] E.J. Coromilas, et al., Worldwide survey of COVID-19–associated arrhythmias, Circ Arrhythmia Electrophysiol. 14 (3) (Mar. 2021) e009458, https://doi.org/10.1161/CIRCEP.120.009458.

[8] N. Hussain, P. Rani, H. Chouhan, U.S. Gaur, Cyber security and privacy of connected and automated vehicles (CAVs)-based federated learning: challenges, opportunities, and open issues, Fed. Learn. IoT Appl. (2022) 169–183.

[9] P. Rani, R. Sharma, Intelligent transportation system performance analysis of indoor and outdoor internet of vehicle (IoV) applications towards 5G, Tsinghua Sci. Technol. 29 (6) (Dec. 2024) 1785–1795, https://doi.org/10.26599/TST.2023.9010119.

[10] F. Colas, et al., FRIPON: a worldwide network to track incoming meteoroids, Astron. Astrophys. 644 (2020) A53.

[11] F. Dai, Y. Shi, N. Meng, L. Wei, Z. Ye, From Bitcoin to cybersecurity: a comparative study of blockchain application and security issues. 2017 4th International Conference on Systems and Informatics (ICSAI), IEEE, Hangzhou, Nov. 2017, pp. 975–979, https://doi.org/10.1109/ICSAI.2017.8248427.

[12] R.W. Ahmad, K. Salah, R. Jayaraman, I. Yaqoob, S. Ellahham, M. Omar, The role of blockchain technology in telehealth and telemedicine, Int. J. Med. Inform. 148 (Apr. 2021) 104399, https://doi.org/10.1016/j.ijmedinf.2021.104399.

[13] I. Abu-elezz, A. Hassan, A. Nazeemudeen, M. Househ, A. Abd-alrazaq, The benefits and threats of blockchain technology in healthcare: a scoping review, Int. J. Med. Inform. 142 (Oct. 2020) 104246, https://doi.org/10.1016/j.ijmedinf.2020.104246.

[14] A. Farouk, A. Alahmadi, S. Ghose, A. Mashatan, Blockchain platform for industrial healthcare: vision and future opportunities, Comput. Commun. 154 (Mar. 2020) 223–235, https://doi.org/10.1016/j.comcom.2020.02.058.

[15] R.M. Aileni, G. Suciu, IoMT: a blockchain perspective (in Studies in Big Data), in: M.A. Khan, M.T. Quasim, F. Algarni, A. Alharthi (Eds.), Decentralised Internet of Things, vol. 71, 71, Springer International Publishing, Cham, 2020, pp. 199–215, https://doi.org/10.1007/978-3-030-38677-1_9 (in Studies in Big Data).
[16] M. Banerjee, J. Lee, K.-K.R. Choo, A blockchain future for internet of things security: a position paper, Digit. Commun. Netw. 4 (3) (Aug. 2018) 149–160, https://doi.org/10.1016/j.dcan.2017.10.006.
[17] M. Aloqaily, I.A. Ridhawi, H.B. Salameh, Y. Jararweh, Data and service management in densely crowded environments: challenges, opportunities, and recent developments, IEEE Commun. Mag. 57 (4) (Apr. 2019) 81–87, https://doi.org/10.1109/MCOM.2019.1800624.
[18] B. Muthu, et al., IOT based wearable sensor for diseases prediction and symptom analysis in healthcare sector, Peer-to-Peer Netw. Appl. 13 (6) (Nov. 2020) 2123–2134, https://doi.org/10.1007/s12083-019-00823-2.
[19] G. Manogaran, N. Chilamkurti, C.-H. Hsu, Emerging trends, issues, and challenges in Internet of Medical Things and wireless networks, Pers. Ubiquit Comput. 22 (5–6) (Oct. 2018) 879–882, https://doi.org/10.1007/s00779-018-1178-6.
[20] H. Abie and I. Balasingham, Risk-based adaptive security for smart IoT in eHealth, in Proceedings of the 7th international conference on body area networks, 2012, pp. 269–275. Accessed: Mar. 03, 2025. [Online]. Available: 〈https://www.researchgate.net/profile/Kshira-Sahoo/post/Is-there-any-work-on-the-risk-assessment-for-IoT-networks/attachment/59d63f0b79197b807799b8d6/AS%3A426142986444802%401478611813459/download/p269-abie.pdf〉.
[21] R.M. Savola, P. Savolainen, A. Evesti, H. Abie, M. Sihvonen, Risk-driven security metrics development for an e-health IoT application. 2015 Information Security for South Africa (ISSA), IEEE, Johannesburg, South Africa, Aug. 2015, pp. 1–6, https://doi.org/10.1109/ISSA.2015.7335061.
[22] M. Mohsin, M.U. Sardar, O. Hasan, Z. Anwar, IoTRiskAnalyzer: a probabilistic model checking based framework for formal risk analytics of the Internet of Things, IEEE Access 5 (2017) 5494–5505, https://doi.org/10.1109/ACCESS.2017.2696031.
[23] B. Zhang, Z. Zou, M. Liu, Evaluation on security system of internet of things based on Fuzzy-AHP method. 2011 International Conference on E-Business and E-Government (ICEE), Shanghai, China, IEEE, May 2011, pp. 1–5, https://doi.org/10.1109/ICEBEG.2011.5881939.
[24] S. Darwish, I. Nouretdinov, S.D. Wolthusen, Towards composable threat assessment for medical IoT (MIoT), Procedia Comput. Sci. 113 (2017) 627–632, https://doi.org/10.1016/j.procs.2017.08.314.
[25] F. Alsubaei, A. Abuhussein, S. Shiva, A framework for ranking IoMT solutions based on measuring security and privacy (in Advances in Intelligent Systems and Computing), in: K. Arai, R. Bhatia, S. Kapoor (Eds.), Proceedings of the Future Technologies Conference (FTC) 2018, vol. 880, 880, Springer International Publishing, Cham, 2019, pp. 205–224, https://doi.org/10.1007/978-3-030-02686-8_17 (in Advances in Intelligent Systems and Computing).
[26] Jun Zhou, Zhenfu Cao, Xiaolei Dong, Xiaodong Lin, "PPDM: a privacy-preserving protocol for cloud-assisted e-healthcare systems, IEEE J. Sel. Top. Signal Process 9 (7) (Oct. 2015) 1332–1344, https://doi.org/10.1109/JSTSP.2015.2427113.
[27] J.J. Hathaliya, S. Tanwar, An exhaustive survey on security and privacy issues in Healthcare 4.0, Comput. Commun. 153 (Mar. 2020) 311–335, https://doi.org/10.1016/j.comcom.2020.02.018.
[28] S. Verma, et al., An automated face mask detection system using transfer learning based neural network to preventing viral infection, Expert Syst. (2024) e13507.
[29] H. Ziglari, A. Negini, Evaluating cloud deployment models based on security in EHR system. 2017 International Conference on Engineering and Technology (ICET), IEEE, Antalya, Aug. 2017, pp. 1–6, https://doi.org/10.1109/ICEngTechnol.2017.8308142.
[30] R. Sanz-Requena, A. Manas-Garcia, J.L. Cabrera-Ayala, G. Garcia-Marti, A cloud-based radiological portal for the patients: IT contributing to position the patient as the central axis of the 21st century healthcare cycles. 2015 IEEE/ACM 1st International Workshop on TEchnical and LEgal aspects of data pRivacy and SEcurity, IEEE, Florence, May 2015, pp. 54–57, https://doi.org/10.1109/TELERISE.2015.18.
[31] A. Singh, et al., Blockchain-based lightweight authentication protocol for next-generation trustworthy internet of vehicles communication, IEEE Trans. Consum. Electron 70 (2) (May 2024) 4898–4907, https://doi.org/10.1109/TCE.2024.3351221.
[32] P. Deshmukh, Design of cloud security in the EHR for Indian healthcare services, J. King Saud. Univ. - Comput. Inf. Sci. 29 (3) (Jul. 2017) 281–287, https://doi.org/10.1016/j.jksuci.2016.01.002.
[33] W. Liu, H. Liu, Y. Wan, H. Kong, H. Ning, The yoking-proof-based authentication protocol for cloud-assisted wearable devices, Pers. Ubiquit Comput. 20 (3) (Jun. 2016) 469–479, https://doi.org/10.1007/s00779-016-0926-8.
[34] C.-M. Chen, S. Liu, X. Li, S.H. Islam, A.K. Das, A provably-secure authenticated key agreement protocol for remote patient monitoring IoMT, J. Syst. Archit. 136 (Mar. 2023) 102831, https://doi.org/10.1016/j.sysarc.2023.102831.
[35] A.K. Idrees, S.K. Idrees, T. Ali-Yahiya, R. Couturier, Multibiosensor data sampling and transmission reduction with decision-making for remote patient monitoring in IoMT networks, IEEE Sens. J. 23 (13) (Jul. 2023) 15140–15152, https://doi.org/10.1109/JSEN.2023.3278497.
[36] N. Shankar, M.K. Nallakaruppan, V. Ravindranath, M. Senthilkumar, B. P. Bhagavath, Smart IoMT framework for supporting UAV systems with AI, Electronics 12 (1) (Dec. 2022) 86, https://doi.org/10.3390/electronics12010086.
[37] G. Ali, M. M. Mijwil, Cybersecurity for sustainable smart healthcare: state of the art, taxonomy, mechanisms, and essential roles, Mesop. J. Cybersecur. 4 (2) (May 2024) 20–62, https://doi.org/10.58496/MJCS/2024/006.
[38] G.R. Pradyumna, R.B. Hegde, K.B. Bommegowda, T. Jan, G.R. Naik, Empowering healthcare with IoMT: evolution, machine learning integration, security, and interoperability challenges, IEEE Access 12 (2024) 20603–20623, https://doi.org/10.1109/ACCESS.2024.3362239.
[39] N. Garg, M. Wazid, J. Singh, D.P. Singh, A.K. Das, Security in IoMT -driven smart healthcare: a comprehensive review and open challenges, Secur. Priv. 5 (5) (Sep. 2022) e235, https://doi.org/10.1002/spy2.235.
[40] Department of Computer Engineering, Faculty of Engineering, Karabuk University, Karabuk, Turkey, et al., Architecture, protocols, and applications of the internet of medical things (IoMT), jcm (2022) 900–918, https://doi.org/10.12720/jcm.17.11.900-918.
[41] B. Cunha, L. Gonçalves, An IoMT architecture for patient rehabilitation based on low-cost hardware and interoperability standards, in: R. Queirós, B. Cunha, X. Fonseca (Eds.), Advances in Medical Technologies and Clinical Practice, IGI Global, 2022, pp. 37–59, https://doi.org/10.4018/978-1-6684-5260-8.ch003.
[42] S. Razdan, S. Sharma, Internet of medical things (IoMT): overview, emerging technologies, and case studies, IETE Tech. Rev. 39 (4) (Jul. 2022) 775–788, https://doi.org/10.1080/02564602.2021.1927863.
[43] Z. Ashfaq, et al., A review of enabling technologies for Internet of Medical Things (IoMT) Ecosystem, Ain Shams Eng. J. 13 (4) (Jun. 2022) 101660, https://doi.org/10.1016/j.asej.2021.101660.
[44] D. Johnson, A. Menezes, S. Vanstone, The elliptic curve digital signature algorithm (ECDSA), IJIS 1 (1) (Aug. 2001) 36–63, https://doi.org/10.1007/s102070100002.